

Practical Exercises #2

1. Configuration of a DHCP Server

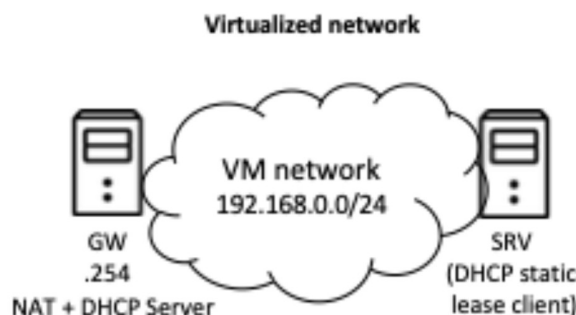
Our goal is to configure a DHCP server in your GW Linux system, **which should be configured to listen on the private network interface**. The first interface is connected to the classroom IPv4 network, while the other should use an IP address from the network 192.168.0.0/24.

The DHCP server should be configured as such:

- Configuration information **common** to all DHCP clients:
 - DNS domain: `siai.dei.uc.pt`
 - DNS servers: 193.136.212.1 and 193.136.212.21
 - NTP servers: `ntp.dei.uc.pt` (there is a small issue, though...)
- For the DHCP clients in the **network 192.168.0.0/24**:
 - Use the address range 192.168.0.100 to 192.168.0.150
 - Identify a client by its MAC address, this host should use the address 192.168.0.111
 - Clients should get from the DHCP server also the netmask address, the broadcast address, and the address of the gateway (192.168.0.254)
 - Client leases should be valid for 8 hours (with a maximum of 10 hours and a minimum of 6 hours)
- The server should ignore requests from the department's LAN. Thus, it must be configured to listen only on the internal network interface.

2. Test your DHCP Server

Test the DHCP server using the SRV Linux system. Configure the interface appropriately.



Goals

Configure a DHCP server in Linux

Test the server using a DHCP client in Linux, Windows or Mac

Materials

- Gestão de Sistemas e Redes em Linux, Jorge Granjal, FCA, “Capítulo 5. Configuração TCP/IP”.
- Gestão de Sistemas e Redes em Linux, Jorge Granjal, FCA, “Capítulo 15. DHCP”.

Other Materials

- [Internet Systems Consortium DHCP KEA](#)